

SSY345 Sensor Fusion HA 3

SHIMING LONG

May 5, 2026

1 Approximations of mean and covariance

(a)

The goal of this task is to provide a numerical "ground truth" for the mean $E[y]$ and covariance $Cov(y)$ of the measurement vector y after the non-linear dual-bearing transformation. The benchmark is established using the Monte-Carlo (MC) method as follows:

1. N_s samples were drawn from the prior state density $p_i(x) = \mathcal{N}(x; \hat{x}_i, P_i)$.
2. Each sample was mapped through the non-linear measurement function $y = h(x) + r$, where $r \sim \mathcal{N}(0, R)$ represents Gaussian measurement noise.
3. The benchmark mean $\hat{y}_{MC}$ and covariance $P_{yy,MC}$ were estimated using sample moments:

$$\hat{y}_{MC} = \frac{1}{N_s} \sum_{j=1}^{N_s} y^{(j)}, \quad P_{yy,MC} = \frac{1}{N_s - 1} \sum_{j=1}^{N_s} (y^{(j)} - \hat{y}_{MC})(y^{(j)} - \hat{y}_{MC})^T \quad (1)$$

For this analysis, we chose $N_s = 10000$ number of samples. **Motivation:** The precision of a Monte-Carlo approximation is governed by the Law of Large Numbers, with the estimation error decreasing at a rate of $1/\sqrt{N_s}$. While $N_s = 100$ is computationally faster, it introduces significant statistical noise, making it unsuitable for a stable benchmark. $N_s = 10000$ ensures that the estimated mean and the 3σ -ellipse accurately capture the non-Gaussian characteristics of the transformed density.

The results of the Monte-Carlo simulation are visualized in Figure 1.

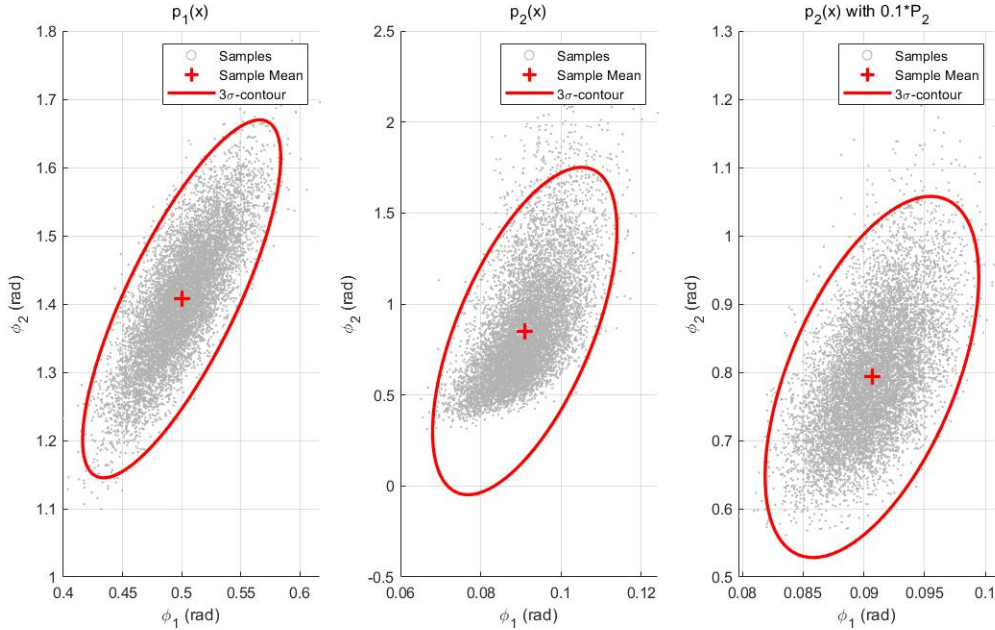


Figure 1: Monte-Carlo samples and 3σ -contours for state densities $p_1(x)$, $p_2(x)$, and $p_2(x)$ with reduced covariance.

For case $p_1(x)$: The samples form a distribution that is nearly ellipsoidal. This indicates that in this region of the state space, the measurement model $h(x)$ is approximately linear.

For case $p_2(x)$: A distinct "banana-shaped" curvature is observed in the sample cloud. The red 3σ -contour, which assumes a Gaussian distribution, fails to perfectly encompass the curved tail of the samples. This highlights the strong non-linearity of the dual-bearing model when the target is closer to the sensors or has higher initial uncertainty.

For case $p_2(x)$ with $0.1P_2$: By reducing the state covariance, the samples occupy a much smaller region in the state space. Consequently, the non-linear transformation $h(x)$ behaves linearly within this local neighborhood, and the resulting measurement density returns to a symmetric Gaussian shape.

(b)

In this task, we compute two different analytical approximations for the mean $E[y]$ and covariance P_{yy} for each of the two state densities, $p_1(x)$ and $p_2(x)$.

The EKF approximates the posterior moments by linearizing the measurement function $h(x)$ using a first-order Taylor series expansion around the prior mean $\hat{x}$.

- **Mean Approximation:** The mean is calculated by passing the prior mean directly through the non-linear measurement model: $\hat{y}_{EKF} = h(\hat{x})$.
- **Covariance Approximation:** The covariance is propagated via the Jacobian matrix $H = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}}$:

$$P_{yy,EKF} = HPH^T + R \quad (2)$$

where P is the prior state covariance and R is the measurement noise covariance.

The UKF avoids explicit linearization by using the Unscented Transform (UT). It represents the state distribution using a set of $2n + 1$ deterministically chosen Sigma points $\mathcal{X}_i$ with associated weights W_i .

- **Propagation:** Each Sigma point is propagated through the non-linear dual-bearing model: $\mathcal{Y}_i = h(\mathcal{X}_i)$.
- **Moment Recombination:** The mean and covariance are recovered from the weighted propagated points:

$$\hat{y}_{UKF} = \sum_{i=0}^{2n} W_i^{(m)} \mathcal{Y}_i, \quad P_{yy,UKF} = \sum_{i=0}^{2n} W_i^{(c)} (\mathcal{Y}_i - \hat{y}_{UKF})(\mathcal{Y}_i - \hat{y}_{UKF})^T + R \quad (3)$$

The computed values for each density are summarized in table 1 . As expected, for $p_1(x)$, both EKF and UKF yield similar results due to low local non-linearity, whereas for $p_2(x)$, the UKF accounts for the non-linear "bias" that the EKF ignores.

Table 1: Numerical comparison of mean approximations for $p_1(x)$ and $p_2(x)$.

State Density	EKF Mean [rad]		UKF Mean [rad]	
	ϕ_1	ϕ_2	ϕ_1	ϕ_2
$p_1(x)$	0.4993	1.4056	0.4999	1.4066
$p_2(x)$	0.0907	0.7854	0.0909	0.8515

(c)

The performance of EKF and UKF approximations is evaluated visually by comparing their 3σ -contours and mean estimates against the Monte-Carlo (MC) benchmark. The results for state densities $p_1(x)$ and $p_2(x)$ are shown in Figure 2 and 3.

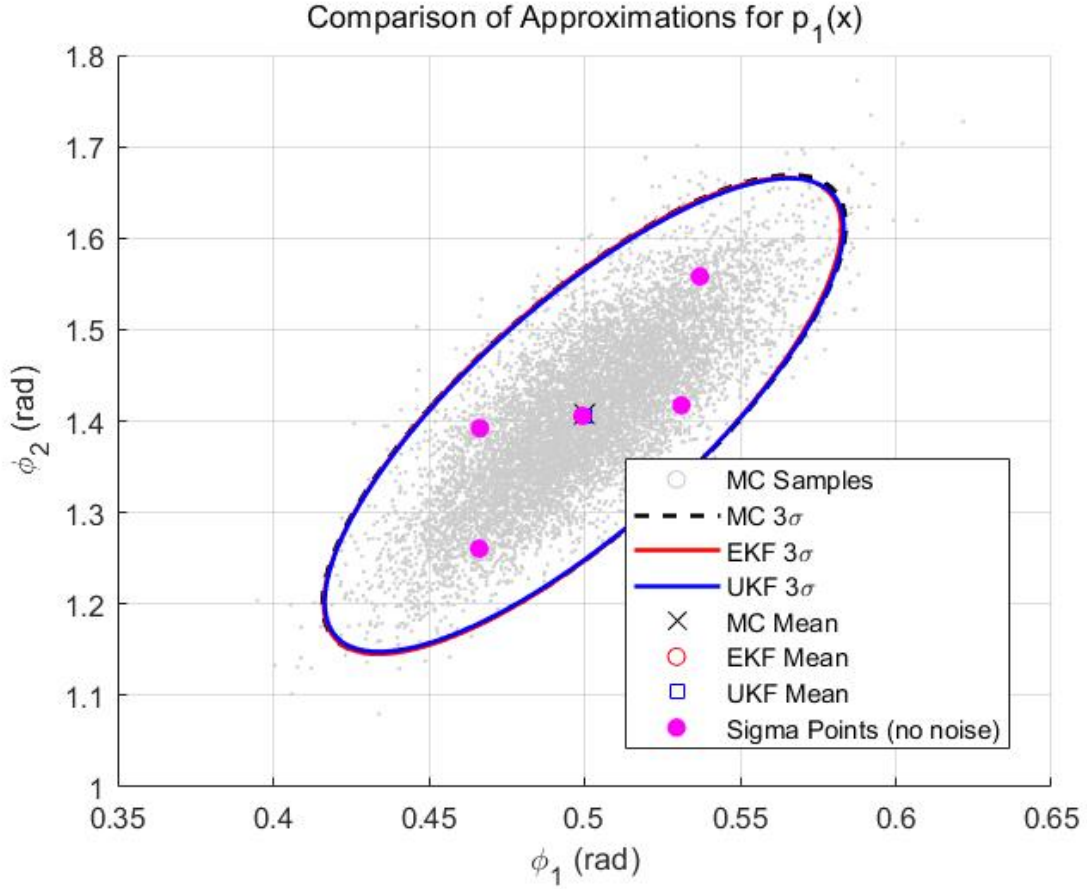


Figure 2: Comparison of approximations for $p_1(x)$

As shown in Figure 2, for the first state density, the measurement model $h(x)$ exhibits near-linear behavior within the uncertainty region.

The 3σ ellipses for MC, EKF, and UKF almost perfectly overlap. All mean estimates coincide at the center of the sample cloud. The propagated sigma points are distributed linearly, consistent with the ellipsoidal shape of the sample distribution.

In contrast, Figure 3 demonstrates a high degree of non-linearity.

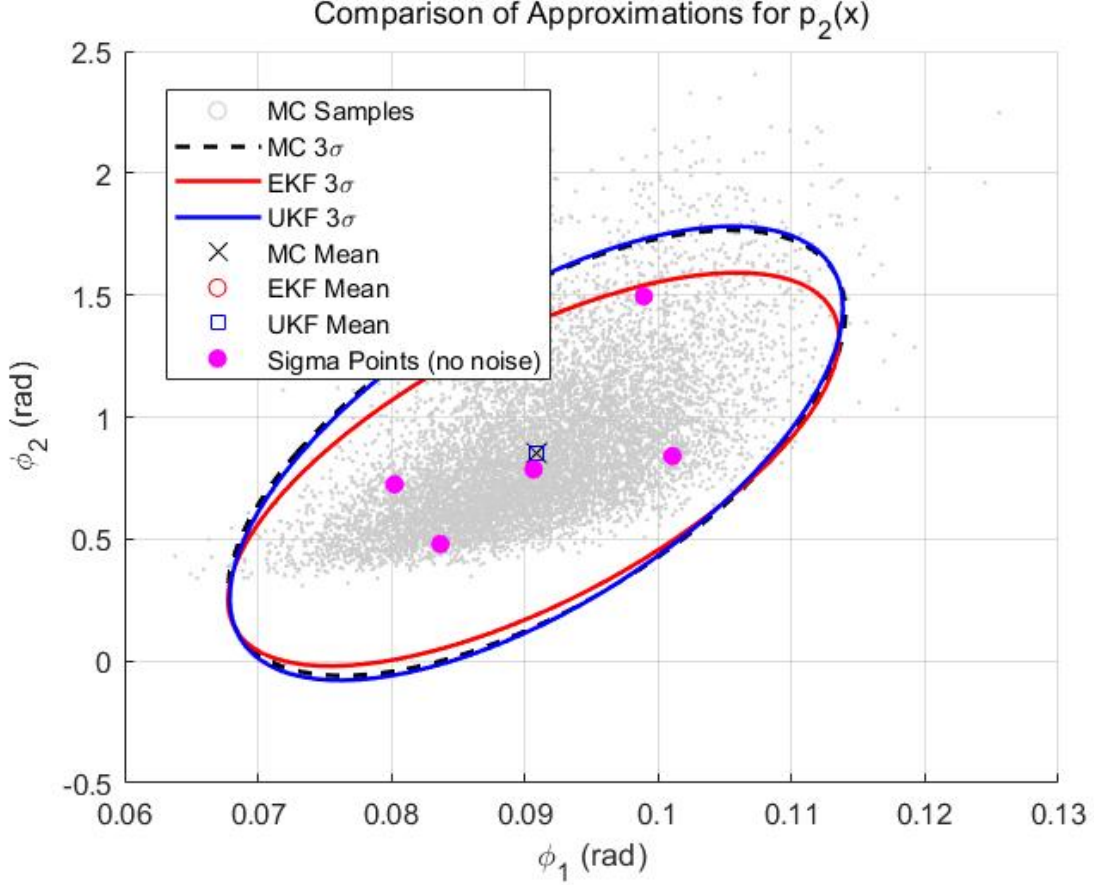


Figure 3: Comparison of approximations for $p_2(x)$

The EKF mean is significantly biased compared to the true MC mean. However, the UKF mean is nearly identical to the MC mean, successfully capturing the "mean shift" caused by the non-linear transformation.

The EKF 3σ ellipse is notably smaller and fails to capture the full spread of the samples, particularly along the ϕ_2 axis. The UKF ellipse aligns much better with the MC benchmark.

The magenta sigma points are distributed along a curve rather than a straight line. This confirms that the UKF "probes" the non-linearity by sampling points across the distribution, whereas EKF only linearizes at a single point.

(d)

By comparing the numerical results and the visualizations, the following differences are observed:

- **Low Non-linearity ($p_1(x)$):** The EKF and UKF yield nearly identical mean estimates. The 3σ -ellipses for both methods, along with the Monte-Carlo benchmark, overlap almost perfectly.
- **High Non-linearity ($p_2(x)$):** A significant discrepancy arises in the second bearing angle ϕ_2 , where the EKF mean is 0.7854 rad and the UKF mean is 0.8515 rad.

Visually, the sample cloud exhibits a "banana-shaped" curvature that the EKF's symmetric red ellipse fails to represent accurately, whereas the blue UKF ellipse aligns closely with the MC benchmark.

The **sigma-point method (UKF)** is superior to the EKF for this dual-bearing measurement model. The EKF relies on a first-order Taylor series expansion, which linearizes the function at a single point (the prior mean) and ignores higher-order terms. In contrast, the UKF uses the Unscented Transform to propagate a set of sigma points through the actual non-linear function.

For $p_2(x)$, the strong non-linearity causes a "mean shift" in the measurement space. The UKF captures this shift and the resulting covariance to at least second-order accuracy, while the EKF remains biased.

If the prior covariance is lowered by a factor of ten, the results for the EKF, UKF, and Monte-Carlo benchmark are expected to converge.

As the state uncertainty shrinks, the samples occupy a much smaller region where the non-linear measurement function $h(x)$ behaves approximately linearly.

In this locally linear regime, the first-order approximation of the EKF becomes highly accurate, and the non-Gaussian "curvature" of the samples disappears. Consequently, both analytical methods will produce nearly identical results to the ground truth.

2 Non-linear Kalman filtering

(a)

In this task, we evaluate the tracking performance of the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF) using a coordinated turn motion model combined with a dual-bearing measurement model.

The simulation was conducted over $N = 100$ time steps with a sampling period of $T = 1$ s. As illustrated in Figure 4, the true trajectory (black solid line) follows the stochastic coordinated turn model. Due to the low measurement noise level ($\sigma_\phi = 0.1^\circ$), the triangulated positions derived from the raw bearing measurements (cyan dotted line) exhibit very low variance and remain highly coincident with the true state throughout the sequence.

A comparison between the EKF (red) and UKF (blue) estimates reveals that both filters provide nearly identical tracking performance in this scenario. This behavior is expected because the target in Case #1 remains at a significant distance from the sensor baseline ($y \approx 200$ m), where the measurement function $h(x)$ exhibits low non-linearity. Consequently, the first-order linearization employed by the EKF is highly accurate, matching the performance of the sigma-point approach used in the UKF.

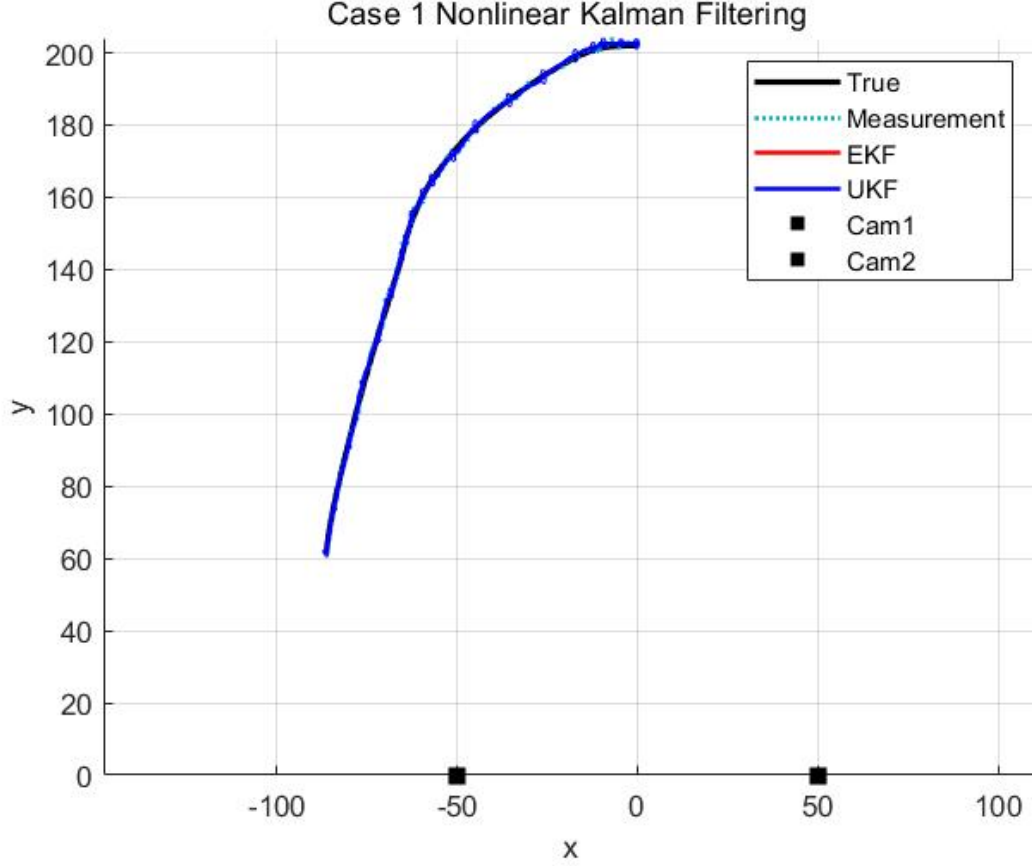


Figure 4: Filtering results for Case 1

To assess filter consistency, 3σ -contours were plotted at every 10th time step. The resulting ellipses are small, reflecting the high confidence of the filters due to the precise measurements. Importantly, the true trajectory stays within the 3σ boundaries across all time steps, demonstrating that both filters are consistent and their covariance matrices $P_{k|k}$ correctly represent the estimation uncertainty.

For the scenario presented in Task 2a, both non-linear filters effectively suppress the process and measurement noise to provide smooth and accurate state estimates. While the UKF is theoretically superior in handling higher-order non-linearities, the geometric configuration of Case #1 allows the EKF to achieve comparable results due to the local linearity of the bearing measurements at long ranges.

(b)

In Task 2b, the non-linear Kalman filters are evaluated under Case 2, utilizing a prior mean of $x_0 = [200, 50, 2, 0, 0]^T$. A defining characteristic shown in Figure 5 is the extreme instability of the triangulated measurements (cyan dotted line) as the target crosses the sensor baseline at $y = 0$. This phenomenon, known as a baseline singularity or geometric dilution of precision, occurs because the target becomes nearly collinear with the sensors at s_1 and s_2 . In this region, the denominator in the intersection formula approaches zero, causing even the minimal bearing noise of 0.1° to produce massive horizontal Cartesian errors.

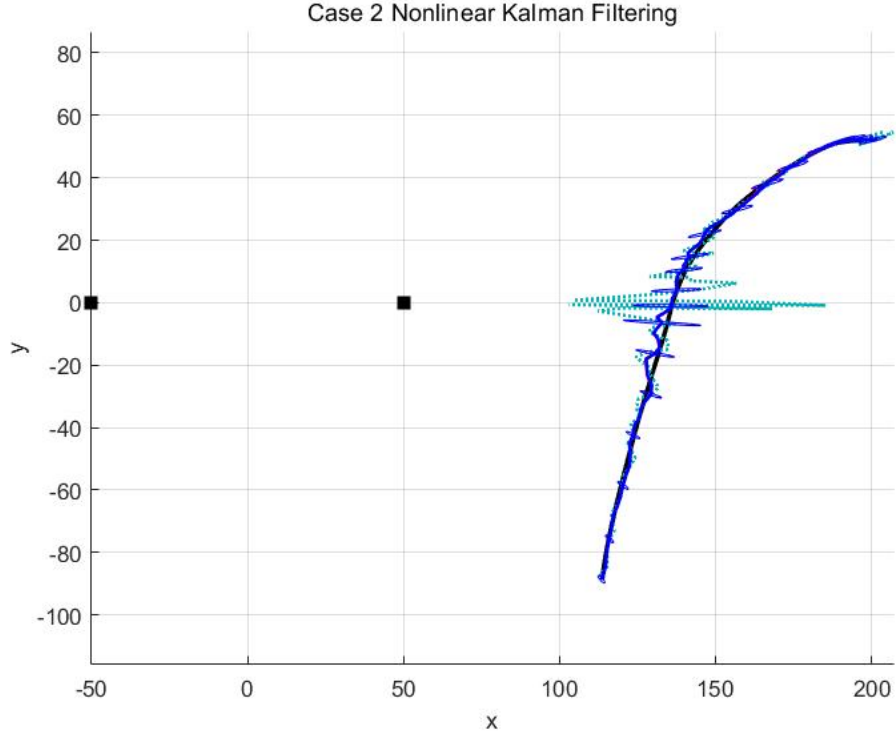


Figure 5: Filtering results for Case 2

Despite the failure of raw triangulation, both the EKF and UKF exhibit robust tracking performance. The filtered estimates (red and blue lines) successfully ignore the erratic measurement outliers by relying on the coordinated turn motion model to predict the state. Because Case 2 involves a trajectory much closer to the sensors than Case 1, the non-linearity of the measurement model $h(x)$ is significantly higher. In this regime, the UKF is theoretically superior as it captures the mean and covariance through sigma-point propagation rather than the first-order Taylor expansion used by the EKF.

The 3σ -contours reorient and expand during the baseline crossing to reflect the increased directional uncertainty. Throughout the simulation, the true trajectory remains within these estimated bounds, which confirms that the filters remain consistent. This demonstrates that the non-linear Kalman filters effectively manage transitions through highly non-linear geometric regions where simple triangulation is insufficient.

(c)

To statistically evaluate the performance of the non-linear filters, a Monte Carlo simulation was performed with 100 independent runs ($MC = 100$) over 100 time steps ($N = 100$), generating 10,000 error samples for each histogram. Figure 6 presents the error distributions for both cases.

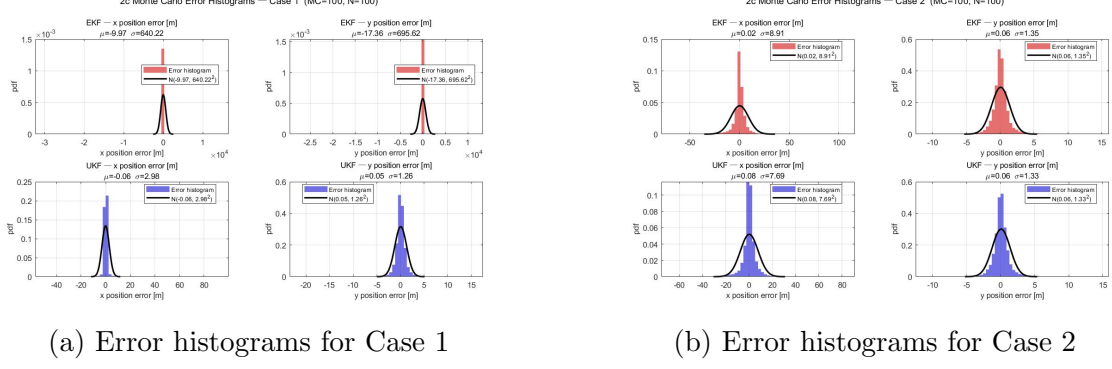


Figure 6: Monte Carlo error histograms for EKF and UKF comparing Case 1 and Case 2.

In Case 1 ($x_0 = [0, 200, 2, 0, 0]^T$), a drastic performance gap is observed. The EKF exhibits extreme standard deviations ($\sigma_x \approx 640$ and $\sigma_y \approx 695$), indicating that the first-order linearization frequently failed or diverged during the 100 runs. In contrast, the UKF remains exceptionally stable with much smaller errors ($\sigma_x \approx 2.98$ and $\sigma_y \approx 1.26$), demonstrating its superior ability to handle initialization at long ranges where the geometry is sensitive to process noise. For Case 2 ($x_0 = [200, 50, 2, 0, 0]^T$), both filters show comparable stability, though the UKF still maintains a slight precision advantage in the x -direction ($\sigma_x \approx 7.69$ for UKF vs 8.91 for EKF).

Regarding the distribution of errors, most successful filtering results (especially the UKF) appear approximately Gaussian, following the bell-shaped probability density function centered near zero. However, we do not strictly expect the errors to be Gaussian. While the Kalman filter framework assumes Gaussian state densities, the non-linear motion model $f(\cdot)$ and measurement model $h(\cdot)$ inherently transform Gaussian inputs into non-Gaussian outputs. The UKF's better fit to the Gaussian PDF is due to its use of the Unscented Transform, which captures higher-order moments that the EKF's linear approximation neglects.

A consistent disparity is noted between the x and y position errors, where the standard deviation for x is significantly larger than for y in all stable runs. This is primarily due to the sensor placement and geometry. The sensors are placed on a horizontal baseline at $s_1 = [-50, 0]^T$ and $s_2 = [50, 0]^T$. For targets located primarily in the $+y$ region, the bearing-only sensors provide better resolution for "depth" (y) than for "lateral" position (x). Furthermore, as the target in Case 2 crosses the $y = 0$ baseline, it encounters a baseline singularity where small angular errors translate into massive horizontal (x) errors due to the ill-conditioned geometry (GDOP), leading to the higher variance observed in the x -direction.

3 Tuning non-linear filters

(a)

The tuning of process noise covariance Q is a critical trade-off between model-based prediction and measurement sensitivity. This task evaluates the EKF's performance quali-

tatively through trajectory tracking and quantitatively using the $\rho(L)$ metric.

As shown in Figures 7a, 7b, and 7c, the choice of Q parameters (σ_v and σ_ω) drastically alters the filter's responsiveness.

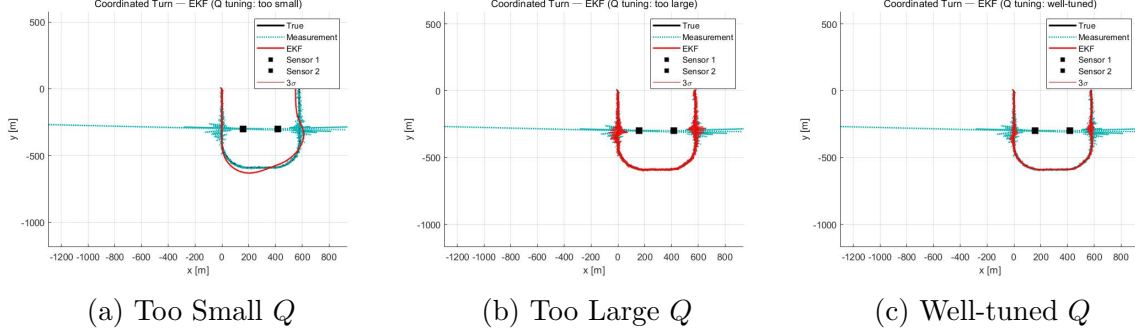


Figure 7: Trajectory tracking results for different process noise settings.

When Q is too small, the filter over-trusts the constant velocity model, leading to significant lag and overshoot during the 90-degree turns. Conversely, a too large Q makes the filter overly sensitive to measurement noise, resulting in a jittery estimated path. The well-tuned case provides a smooth trajectory that accurately tracks maneuvers with minimal delay.

(b)

The normalized estimation error $\rho(L)$ is utilized to evaluate the filter's consistency. The target value for a consistent filter is $\rho(L) \approx 1$.

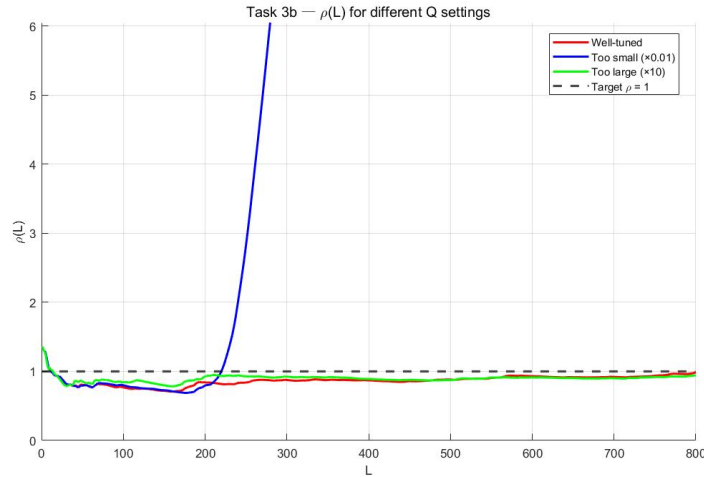


Figure 8: Normalized estimation error $\rho(L)$ for different Q settings.

The analysis of Figure 8 yields the following observations:

1. Too Small Q (Blue): $\rho(L)$ remains low on straight segments but spikes exponentially (> 6) during maneuvers. This indicates an overconfident and inconsistent filter that fails to account for the model errors during turns.

2. Too Large Q (Green): $\rho(L)$ stays consistently below 1.0, indicating a **conservative** filter that overestimates its error covariance, sacrificing precision for noise sensitivity.
3. Well-tuned Q (Red): The metric converges and fluctuates near 1.0, confirming that the filter is consistent and the estimated $P_{k|k}$ accurately reflects the true error variance.

(c)

Figure9 shows the EKF estimates of speed, heading and turn-rate compared to the true values, using the well-tuned process noise covariance $\mathbf{Q}$.

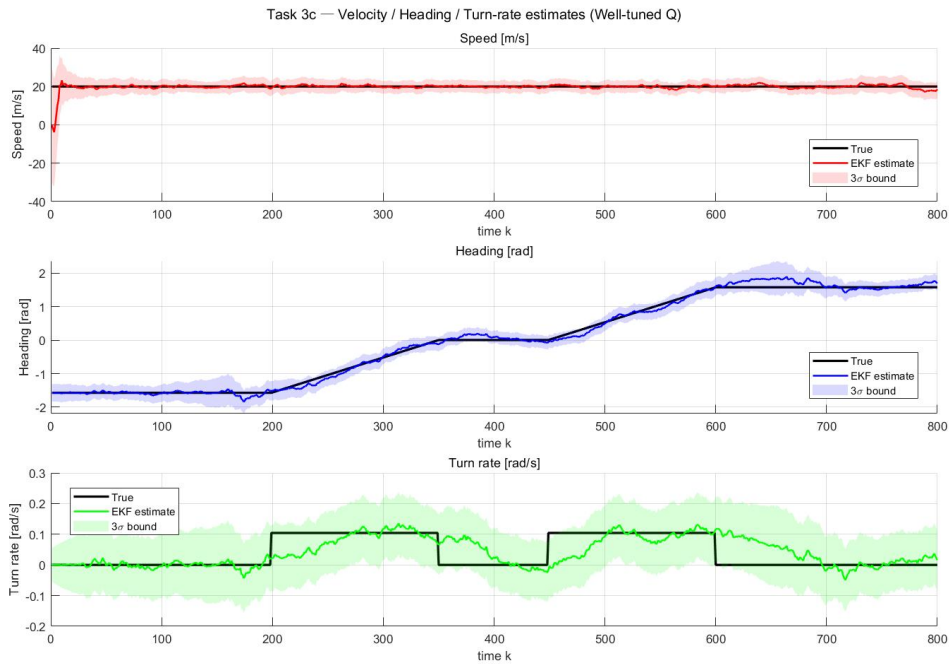


Figure 9: EKF estimates of speed, heading and turn-rate with well-tuned $\mathbf{Q}$

From Figure9, the following observations can be made:

1. The filter converges quickly from the uninformative prior and tracks the true speed of 20 m/s accurately throughout the entire sequence, including during the turns. The 3σ bound is narrow after the initial transient, indicating high confidence in the speed estimate.
2. The heading estimate follows the true value well during straight segments. However, noticeable deviations appear at the transitions into and out of each turn (around $k = 200$, $k = 350$, $k = 450$, $k = 600$), where the heading changes rapidly. The 3σ bound widens at these transitions, reflecting increased uncertainty.
3. During straight segments the true turn-rate is zero, and the filter correctly estimates values close to zero. During the turning segments, the estimated turn-rate tracks the true value approximately but with considerable noise and a wide 3σ band. This is

because the turn-rate enters the measurement only indirectly through the nonlinear motion model, making it poorly observable from bearing-only measurements.

In summary, it is not possible to achieve uniformly accurate estimates of all three states throughout the entire sequence with a fixed $\mathbf{Q}$. Speed can be estimated well, heading reasonably well, but turn-rate remains noisy — especially at the transitions.

Figure10 shows the position RMSE broken down by segment.

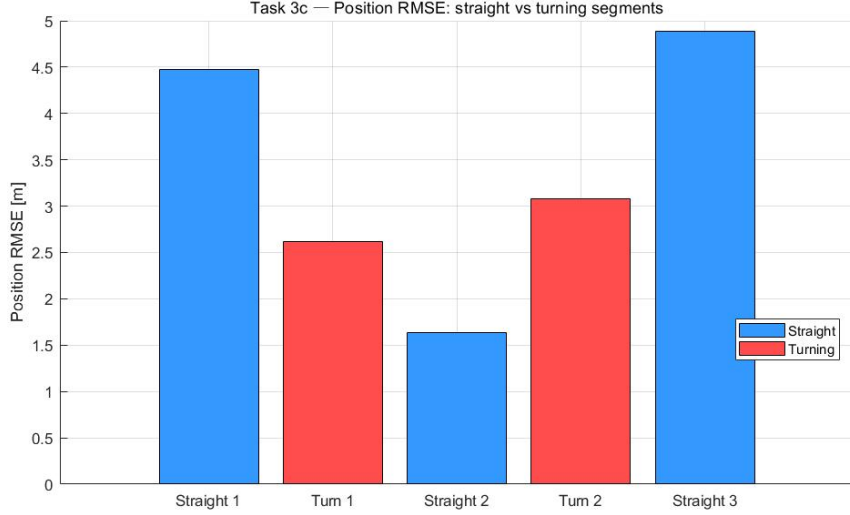


Figure 10: Position RMSE per trajectory segment with well-tuned $\mathbf{Q}$

A clear conflict exists between the desired $\mathbf{Q}$ tuning for the two types of segment:

1. During straight segments, the true turn-rate is exactly zero and the motion is nearly constant velocity. A *small* process noise is preferred here, so that the filter is confident in its predictions and does not add unnecessary uncertainty. However, as seen in Figure10, Straight 1 and Straight 3 exhibit relatively high RMSE (≈ 4.5 m and ≈ 4.9 m respectively). This is because the filter has not yet fully converged at the start, and after the second turn it has accumulated heading uncertainty.
2. During turning segments, the turn-rate changes abruptly from zero to $\omega = \pi/301T$ rad/s. A large process noise on ω is needed so the filter can react quickly to this sudden change. With a small $\mathbf{Q}$, the filter is too rigid and lags behind the turn, causing large position errors. With a large $\mathbf{Q}$, the filter reacts fast but introduces unnecessary noise during straight segments.
3. At the transitions, the mismatch is most severe. The true trajectory is generated by an *instantaneous* step change in turn-rate (as set by `omega(200:350)`), which is a model mismatch relative to the smooth coordinated-turn motion model used in the filter. No fixed $\mathbf{Q}$ can handle both the rigidity required on straight segments and the agility required at transitions simultaneously.